Skills Summary

Derivative Applications and Modeling

Use this reference while completing your worksheet or when studying.

Which kind of problem is this? Read the question before touching the algebra — all three types differentiate, and they differ in what you differentiate *with respect to*.

- **Related rates:** two quantities change in time. Differentiate the relation with respect to t, keeping every changing letter.
- **Optimization:** one quantity is largest or smallest. Reduce to one variable, set the derivative to zero, then check the domain.
- **Curve analysis:** the shape of a graph. f' gives increasing/decreasing and critical points; f'' gives concavity and inflection points.

1. Curve analysis with f' and f''

Critical numbers: where $f'(x) = 0$ or f' is undefined. $f' > 0$ means f increases; $f' < 0$ means it decreases.

$f'' > 0$ is concave up, $f'' < 0$ concave down, and an inflection point needs f'' to *change sign* — $f''(x) = 0$ by itself is not enough ($g(x) = x^4$ at the origin is the standard counterexample).

Example: Find the critical numbers of $f(x) = 2x^3 - 9x^2 + 12x$.
Step 1. The question is about the shape of the graph, so the tool is f', and critical numbers are exactly its zeros for a polynomial.
Step 2. $f'(x) = 6x^2 - 18x + 12 = 6(x-1)(x-2)$, which is zero when a factor is zero.
$\Rightarrow$ $\boldsymbol{x = 1}$ and $\boldsymbol{x = 2}$ (f' is $+$, then $-$, then $+$: a maximum at 1, a minimum at 2)

Try it: Find the critical numbers of $g(x) = x^3 - 12x + 5$.

check: $\boldsymbol{x = -2}$ and $\boldsymbol{x = 2}$

2. Related rates

Order matters. Write the geometric relation, use any fixed ratio to remove extra variables, differentiate with respect to t, and only *then* substitute the instant's numbers. Substituting first freezes a variable that is still moving.

Circle: $A = \pi r^2 \Rightarrow \frac{dA}{dt} = 2\pi r\,\frac{dr}{dt}$. Sphere: $V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2\,\frac{dr}{dt}$.

Example: A circular ripple's radius grows at 3 cm/s. How fast is its area growing when $r = 4$ cm?
Step 1. Two quantities change with time and the link between them is the area formula, so differentiate that link in t before any number goes in.
Step 2. $\frac{dA}{dt} = 2\pi r\,\frac{dr}{dt} = 2\pi(4)(3)$.
$\Rightarrow$ $\boldsymbol{24\pi}$ square centimetres per second

Try it: A square's side grows at 5 in/s. How fast is its area growing when the side is 7 in? (Use $\frac{dA}{dt} = 2s\,\frac{ds}{dt}$.)

check: **70**

3. Optimization with a domain check

Name the quantity to be optimized, write it in *one* variable using the constraint, differentiate, and solve. Then finish the job: list the critical numbers *inside* the physical domain plus the endpoints, and compare — or give a sign chart for the derivative. A root outside the domain is discarded with a reason, not ignored.

Example: 40 m of fence, three sides of a rectangle against a wall. Which x (the two equal sides) maximizes the area $A(x) = x(40 - 2x)$?

Step 1. The area is already a function of the single variable x on $0 < x < 20$, so the maximum can only be at a zero of A' or at an endpoint.

Step 2. $A(x) = 40x - 2x^2$, so $A'(x) = 40 - 4x = 0$.

$\Rightarrow$ $\boldsymbol{x = 10}$ ($A'' = -4 < 0$, so this critical point is the absolute maximum, and both endpoints give area 0)

Try it: Same fence problem with 24 m of fence: $A(x) = x(24 - 2x)$. Which x maximizes the area?

check: $\boldsymbol{x = 6}$

(!) Watch out: the two most expensive habits on this topic are substituting the instant's numbers into a related-rates relation *before* differentiating, and stopping an optimization at "$f'(x) = 0$ here" without saying why that point is the absolute extreme on the domain. Both lose points even when the number is right.